

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Detailed Marking Report — Chemistry Exam-22

Seemab · Class 9 (Science Group) · Revision Test, Chapters 1 to 9 · 50-mark school-mirror paper · Attempted 14 September 2026

42 / 50

TOTAL SCORE

84.0%

PERCENTAGE

A

GRADE

9 / 10

SECTION A
90%

20 / 24

SECTION B
83.3%

13 / 16

SECTION C
81.3%

The short version

Every one of the three big errors from her last two chemistry papers came back fixed: she named the oxidizing and reducing agents correctly, she stated that a catalyst does **not** change ΔH , and she got four of her five exam-21 MCQ misses right (Bohr vs orbitals, electron affinity, giant covalent, and the neutron/proton inverse). Her MCQ score is 9/10, though on a ten-question section that is a narrow sample.

What cost marks tonight is different: **she stopped short of the full instruction** in three places (no data quoted in 2(i)(c), no state symbols in 3(ii)(e), no catalysed curve in the energy diagram), and two **careless slips** on work she otherwise understands (a sign slip giving $N = +3$ in NH_3 , and writing 6.98×10^{23} g where the answer is 6.97×10^{-23} g). The one true content gap still standing is the **catalyst-to-process list**: she picked platinum for margarine.

How this paper was marked

The scan is a 9-page CamScanner PDF filed at attempts/2026-09-14/scans.pdf. All nine pages were read at 110 DPI, and every item that decides a mark was re-read at 400 DPI with contrast enhancement: the MCQ list, the X group line in 2(iii), the HCl Lewis structure, the NH_3 oxidation-number working, the energy diagram, and the exponent in 3(ii)(f). The ten correct MCQ answers were re-derived independently from the rendered exam paper rather than trusted from the key; all ten keys are correct and her letters are unambiguous. She wrote her MCQ answers as a numbered list on page 1, with a written reason beside Q1–Q5. All sections and all sub-parts are present in order; no page is missing.

Five marks rest on judgement calls rather than clear right/wrong reads. They are listed at the end of this report so Atif can rule on each one. The total can move between 40.5 and 45 depending on those rulings.

SECTION A — Multiple Choice (9 / 10)

#	Topic	Hers	Key	Mark	Comment
1	Technology defined	B	B	1	Correct. She also wrote a reason distinguishing science, engineering and technology. Ch1 §1.5.
2	Bonds per carbon in diamond	D	D	1	Correct, four in a tetrahedral network. Reason given: “other options just doesn’t make any sense”. Ch2 §2.3.
3	Bohr-only statement	A	A	1	Correct, fixed circular orbits. Re-test of her exam-21 Q9 miss (she gave Bohr for the QM model), now fixed. Her reason: “the next options also linked with quantum mechanical model”, which is exactly the right elimination.
4	Neutron emission	C	C	1	Correct, another isotope. Inverse of her exam-20 Q14 proton miss. Ch3 §3.3.
5	Electron affinity defined	A	A	1	Correct. Re-test of her exam-21 Q13 miss (ionization energy’s definition), now fixed. Her reason explicitly rules out ionization energy, electronegativity and shielding effect.
6	Catalyst for margarine	C	B	0	She chose platinum . Nickel is the margarine catalyst (iron = Haber, platinum/palladium = catalytic converters, platinum also Contact process). Re-test of her exam-21 Q14 miss (she said nickel for Haber), still not fixed: the catalyst-to-process list is the one confusable set that has now cost her on two consecutive chemistry papers. No reason written for Q6–10. Ch4 §4.5.
7	Why SiO_2 is hard	D	D	1	Correct, giant covalent. Re-test of exam-21 Q19 miss, now fixed. Ch5 §5.5.
8	Subscript outside a bracket	B	B	1	Correct, two ammonium ions. Ch6 §6.3.
9	Removal of hydrogen	C	C	1	Correct, oxidation. Ch7 Key Points.
10	$CoCl_2$ with water	A	A	1	Correct, pink. Ch9 §9.3.

SECTION B — Short Questions (20 / 24)

Q	Task	Awarded	Comment (her words quoted)
2(i)	Saturated / unsaturated; temperature effect from Table 2.2	3 / 4	(a) “A saturated solution is a solution that contains the maximum amount of solute that can dissolve in a given amount of solvent at a particular temperature”: 1. (b) “...contains less than the maximum amount of solute that can dissolve at a particular temperature”: 1. (c) “Increasing temperature increase the solubility of NH_4Cl , but decreases the solubility of $Ca(OH)_2$ ”: both directions correct, but the question said “Using the data” and she quoted

			no figure from the table. Key: direction with no data = ½ each, so 1 / 2 . <i>Judgement call 1</i> .
2(ii)	Relative atomic mass of bromine; p/n/e; chemical properties	4 / 4	(a) $(79 \times 50.5) + (81 \times 49.5)$ over 100 = $(3989.5 + 4009.5)/100 = 7999/100 = 79.99$, full working: 2 . (b) protons 35, electrons 35, neutrons $81 - 35 = 46$: 1 . (c) “They have the same number and arrangement of electrons, especially the same number of outer shell electrons”: 1 .
2(iii)	Period and group of X ($3s^2$) and Y ($3s^23p^4$); ions; formula	3 / 4	(a) X: “highest shell = 3 → period 3rd”, then “outer electrons = 2 → group: $8+10 = 18$ ”. She struck out the “+10” and the “1”, which leaves Group 8 , not Group 2. (The “2” for outer electrons also looks written over another digit.) So she knew the +10 did not belong on an s-block element, but the number left behind is still wrong. Y: “Period 3rd, group $6+10 = 16$ ”, correct. Key awards each element’s mark only with both period and group, so X 0 , Y 1 . <i>Judgement call 2</i> (½ for X’s period is defensible). (b) “X has 2 outer electrons, so it loses 2: X^{2+} ”; “Y needs 2 electrons to complete its outer shell: Y^{2-} ”: 1 . (c) XY with charges +2 and –2 marked: 1 .
2(iv)	HCl dot-and-cross; polarity	3 / 4	(a) Her first attempt is a Lewis structure: $H \times \bullet Cl$ with dot pairs above, to the right and below Cl, plus a fourth pair drawn further below . Two circle-style diagrams after it are scribbled out. The shared pair is correct (1), but the drawing shows four lone pairs on chlorine (ten electrons) instead of three (eight), so the octet mark is not earned: 1 / 2 . At 400 DPI the fourth pair is clearly separate ink, not a re-trace, but it is small. <i>Judgement call 3</i> . (b) “The H–Cl bond is polar because Cl is more electronegative than H, so the bonding electrons are attracted more strongly towards chlorine”: 2 .
2(v)	Redox in $2NH_3 + 3CuO \rightarrow 3Cu + N_2 + 3H_2O$	3 / 4	(a) CuO: $x + (-2) = 0$, $x = +2$; Cu = 0: 1 . (b) NH_3 : she wrote “ $x + 3(1) = 0$, $x - 3 = 0$, $x = +3$ ”. A sign slip in the second line: it should be $x + 3 = 0$, $x = -3$. $N_2 = 0$ is correct (after two scribbled attempts). ½. (c) “Nitrogen $+3 \rightarrow 0$ {oxidized}; copper $+2 \rightarrow 0$ {reduced}”. The identification is correct , but with her own $+3 \rightarrow 0$ the oxidation number goes down, which would mean reduction, so the reasoning contradicts the label. ½. (d) “oxidizing agent = CuO, reducing agent = NH_3 ”, whole molecules, not inverted: 1 . This is the exact exam-20 error (agents inverted) and it is fixed.
2(vi)	$2SO_2 + O_2 \rightleftharpoons 2SO_3$ sealed container	4 / 4	(a) forward and reverse equations both written: 1 . (b) “At the very start no SO_3 has been formed, so no SO_3 available to react in the reverse direction. Therefore reverse reaction rate = 0”: 1 . (c) rate of forward = rate of reverse; concentrations of reactants and products remain constant: 2 .

SECTION C — Long Questions (13 / 16)

Q	Task	Awarded	Comment (her words quoted)
3(i)	$N_2 + O_2 \rightarrow 2NO$: bond energies, energy diagram, catalyst	6.5 / 8	(a) $[945 + 498] = 1443$; $2(630) = 1260$; $1443 - 1260 = +183 \text{ kJ/mol}$: 3 . (b) “Endothermic reaction: ΔH is positive, meaning more energy is [needed to] break bonds than is released when new bonds form”: 1 . (c) Axes labelled energy, reactants low, products higher, ΔH labelled $+183 \text{ kJ/mol}$: 1 . One peak labelled “without catalyst”, but “activation energy” is never written: ½. There is an extra small bump on the rising line, but it is unlabelled and on the same path, so it is not a catalysed curve, and no lower catalysed hump is drawn: 0 . Diagram 1.5 / 3 . <i>Judgement call 4</i> . (d) “Catalyst changes: activation energy, it lowers E_a . Does not change: enthalpy change ΔH ”: 1 . This is the exam-21 Q27 misconception (catalyst changes ΔH), and it is fixed.
3(ii)	Hydrocarbon P: EF, MF, moles, molecules, combustion, mass of one molecule	6.5 / 8	(a) $n(C) = 1.8/12 = 0.15 \text{ mol}$; $n(H) = 0.3/1 = 0.30 \text{ mol}$; divide by smallest, 1 : 2, CH_2 : 2 . (b) EF mass $12 + 2 = 14$; $42/14 = 3$; $(CH_2)_3 = C_3H_6$: 1 . (c) $8.4/42 = 0.20 \text{ mol}$: 1 . (d) $0.20 \times 6.022 \times 10^{23} = 1.204 \times 10^{23}$ molecules: 1 . (e) She worked the balancing step by step in three lines, noted “there are 9 O atoms on the right, multiply by 2”, and reached $2C_3H_6 + 9O_2 \rightarrow 6CO_2 + 6H_2O$, perfectly balanced, but with no state symbols , which the question asked for in bold: 1 / 2 . (f) Working $42 / 6.022 \times 10^{23}$ is correct, but the final answer is written “ $6.98 \times 10^{23} \text{ g}$ ”: the minus sign on the exponent is missing (should be $6.97 \times 10^{-23} \text{ g}$). Method right, answer off by 46 powers of ten: ½. <i>Judgement call 5</i> .

The re-tests this paper was built to run

Earlier error	Re-test on exam-22	Result
Oxidizing and reducing agent inverted (exam-20 Q55/57)	2(v)(d)	FIXED
“A catalyst changes ΔH ” (exam-21 Q27)	3(i)(d)	FIXED
Bohr model given for the QM model (exam-21 Q9)	MCQ 3	FIXED
Ionization energy given for electron affinity (exam-21 Q13)	MCQ 5	FIXED
Simple molecular given for giant covalent (exam-21 Q19)	MCQ 7	FIXED
Proton emission → isotope (exam-20 Q14)	MCQ 4 (inverse)	FIXED
Nickel given for the Haber catalyst (exam-21 Q14)	MCQ 6 (margarine)	NOT FIXED
Group rule vs period rule (exam-20 Q21/Q30)	2(iii)(a)	HALF

Six of nine fixed, one half-fixed, two not. The group/period result is the interesting one: for the s-block element X she wrote “8 + 10 = 18”, then struck out the +10 and the 1, leaving Group 8 instead of Group 2. So she now knows the +10 does not apply to an s-block element, but she is still counting the wrong number of electrons for its group. She applied +10 correctly to the p-block element Y.

What to tell her

- 1. Learn the catalyst list as one table:** iron → Haber (ammonia); nickel → margarine; platinum and palladium → catalytic converters; platinum → Contact process. It has now cost her a mark on two chemistry papers in a row.
- 2. State symbols, again.** This is the one habit that has survived every other fix. When a question says “with state symbols”, write (g), (l), (s), (aq) before moving on.
- 3. “Using the data” means quote a number.** Her chemistry in 2(i)(c) was right; the missing figures cost the mark.
- 4. Check signs when solving for an oxidation number:** $x + 3(+1) = 0$ gives $x = -3$. Then check the direction matches the word (up = oxidized, down = reduced). And check the sign of a power of ten: the mass of one molecule is tiny, so the power must be negative.
- 5. An energy diagram question lists what to label; tick each label off.** She drew a good uncatalysed curve but never wrote “activation energy” and never drew the catalysed hump.

Judgement calls for Atif to rule on

#	Item	Awarded	Range
1	2(i)(c) directions correct, no figures quoted despite “Using the data”	1 / 2	1 to 2
2	2(iii)(a) X period correct, group corrected from 18 to 8 (should be 2)	1 / 2	1 to 1.5
3	2(iv)(a) HCl Lewis structure with four lone pairs on Cl instead of three	1 / 2	1 to 2
4	3(i)(c) energy diagram: no “Ea” label, no catalysed curve	1.5 / 3	1 to 1.5
5	3(ii)(f) correct working, final exponent missing its minus sign	½ / 1	0 to 1

Most generous reading: 45/50. Strictest: 40.5/50. **42/50 is the key applied as written.**